

Latest Display Modules

Module	Manufacturer	Resolution	Dimensions (WxHxD mm)	Brightness (cd/m ²)	Interface	Contrast ratio	Viewing angle (U/D/L/R)	Type
DET035QVNTCM1-1W	Densitron	320x240	76.9x63.9x4.85	500	I2C	800:1	80/80/80/80	Multi-touch TFT module
WF62A	Winstar	640x320	170.32x88.3x5.3	1000	MCU	800:1	80/80/80/80	TFT module
AA070ME11	Mitsubishi Electric	800x480	118.5x84.7x3.9	1500	LVDS	600:1	60/80/80/80	TFT-LCD module
AA050MG04	Mitsubishi Electric	800x480	118.5x84.7x3.9	1000	CMOS	1000:1	85/85/85/85	TFT-LCD module
TX172D00VMOBAA	KOE	800x480	154.8x92.7x10.2	700	LVDS	600:1	85/85/85/85	TFT module
AMS465GS45	Samsung	720x1280		300	LVDS	3400:1	80/80/80/80	OLED module
H499TLB01 V0	AUO	720x1280		250	MIPI	1000:1	80/80/80/80	OLED module
DLA320480AB035F	Densitron	320xRGBx480	52.9x83.5x1.22	180	MIPI-DBI 8/16/24 bit MPU/RGB, MIPI-DBI Serial			OLED module

TFTs with enhanced optics

Displays combining exceptional projected-capacitive-touch (PCT) technology with industrial-grade TFTs are able to offer higher transmissivity and unlimited controller-dependent multi-touch capability. Now we have a combination of high resolutions with extremely compact form factors up to depths as low as 4.6mm, enabling the design of even thinner devices or providing space for integrating more components.

Modules like those in Densitron's DET035QVN series have improved TFTs with ultra-wide-view (UWV) polariser solution, increased viewing angles in each direction with greatest colour consistency, which is well suited for use in high-end, graphics-rich applications, for both fixed and mobile devices. And they consume as low as 2.8W.

If we talk about mono TFTs, the latest monochrome models with TN-positive LCD applied in the active-matrix TFTs are able to provide brightness as high as 1000-1500 nits, and high contrasts as much as 800:1 with short response times. These models, like the Winstar's Mono WF57S, are superior alternatives to traditional monochrome LCD modules.

Thin-glass substrate EPD. Based on new thin-glass TFT technology, the latest E-Ink or electronic paper display (EPD) modules are catering to the demands of portable consumer electronics products that are much lighter and thinner as compared to other standard displays. These weigh around 50 per cent lighter than an equivalent glass-based TFT and are 50 per cent thinner.

Reflective displays for lower-power applications

For any embedded application, primary concern is power. Most displays use a backlight that is projected through the display. This backlight consumes most of the power. Therefore eliminating the need for any backlight in displays will surely increase the battery life of the products.

In contrast to the backlight technology, we have power-efficient reflective-technology-based E-Ink displays. These do not require any kind of backlight and take ambient light from the environment, which is reflected from the surface of the display, thereby saving a huge amount of power.

E-Ink: What adds to the power efficiency

Another advantage of E-Ink displays, which adds to the power efficiency of their products, is the bistability of an EPD. An image on the E-Ink display screen is retained even if all the power sources are removed. Thus, these display modules could be used to make the most power-efficient displays that will consume power only when something is changing and not when it is idle.

No idle refreshing is required in E-Ink displays, while any traditional LCD display needs to be refreshed around 30 times per second, regardless of whether or not anything new is being displayed.

In addition to being thin and light, these modules deliver the same energy efficiency and daylight readability of all E-Ink displays. Besides, the energy efficiency further reduces weight of the end products by requiring a much smaller battery.

Flexible active-matrix displays for wearable electronics. MOBIUS is another E-Ink technology that is mainly used by smartwatch start-ups like Sonostar and Transmart for designing flexible wearable display solutions. MOBIUS-based modules are smaller version of displays based on flexible-TFT technology. Lightweight and rugged properties make them conformable, so the end product has a better fit for the consumer.

With the inherent power management capability, these EPD modules are ideal for lightweight portable products that require a large display surface. Also, if you

only want to display some numbers, and those numbers change very infrequently, no point in refreshing the display every time to burn power.

OLED technology for high-end mobile devices

OLED technology has been around for quite some time, but now it is evolving with more advanced features and physics, promising to be the technology for the future designs. OLED displays are improving at a very impressive rate of about 20 per cent or more year-on-year in brightness, colour management, colour accuracy, resolution, PPI and power efficiency.

OLED technology has many advantages over the other display technologies in terms of performance as well as power. Avinash Babu, senior architect, Mistral Solutions, explains, "People are slowly moving the display